Bootstrap for Regression: The Basic Idea (Lab)

Consider the linear regression model

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

The least-squares estimator is $\widehat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}, \qquad \boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}.$

$$X = \begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{p1} \\ 1 & x_{12} & x_{22} & & x_{p2} \\ \vdots & & & & \\ 1 & x_{1n} & x_{2n} & & x_{pn} \end{bmatrix}$$

$$\hat{\beta} \sim N\left(\beta, \sigma^2 (X^T X)^{-1}\right) \Longleftarrow \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \underline{\beta} = (\beta_0, \beta_1 \cdots \beta_p)$$

under the assumption that $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

Tricky $\Longleftrightarrow$ we dont know the distn of ε_i's

then we cant say $\hat{\beta} \sim N(\,)$

no closed sampling distn $\rightarrow$

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Why bootstrap regression?

The sampling distribution of $\widehat{\boldsymbol{\beta}}$ may be difficult to derive exactly, particularly when the error distribution is unknown. Bootstrap resampling approximates this distribution using the observed data.

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Bootstrap principle (Recall)

1. Treat the observed sample as an approximation to the population.
2. Generate B bootstrap datasets.
3. Refit the regression in every dataset.
4. Use the resulting estimates to quantify uncertainty.

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What do we obtain?

$\widehat{\beta}_j^{*(1)}, \ldots, \widehat{\beta}_j^{*(B)}$

- Bootstrap standard error
- Confidence intervals
- Sampling-distribution plots
- Assessment of estimator stability

$$\widehat{\mathrm{SE}}_{\mathrm{boot}}(\widehat{\beta}_j) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\widehat{\beta}_j^{*(b)} - \overline{\beta}_j^{*})^2}$$

Pairs (Case) Bootstrap: Resample Observations

Each observational unit is the complete pair $(\vec{x}_i, y_i)$.

$\underline{x}_i = (1, x_{i1} \ldots x_{ip})$

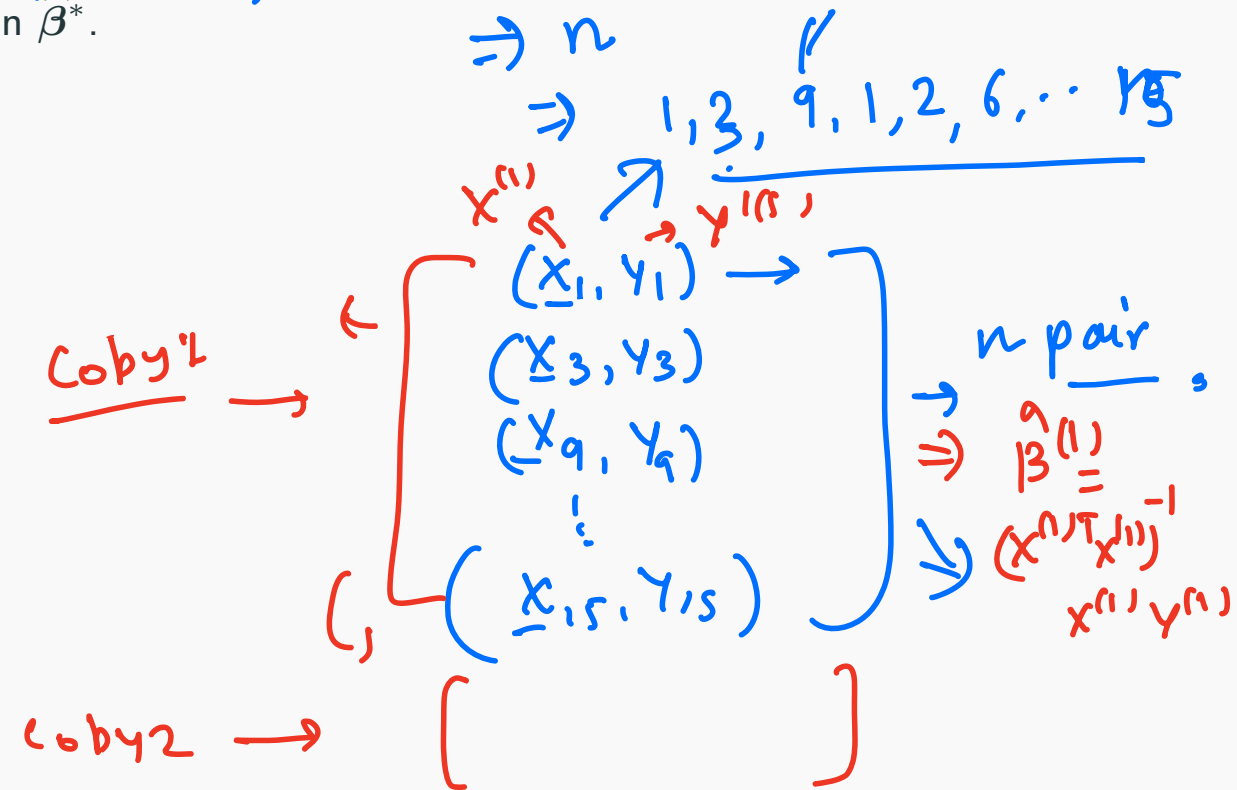
↑ input ; independent, or Covariates

response output

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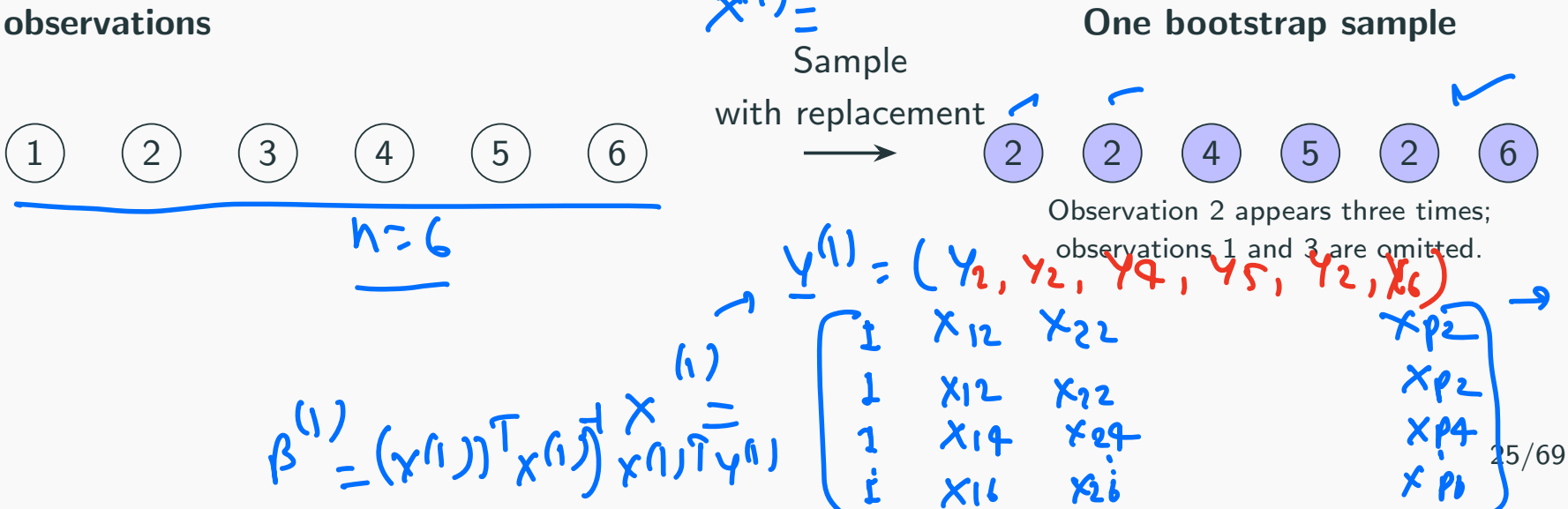
1. Draw n indices independently with replacement from $\{1, \ldots, n\}$.
2. Form the bootstrap sample $\{(x_{i_k^*}, y_{i_k^*})\}_{k=1}^{n}$.
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Key implication

Both predictors and responses are resampled together. Therefore, the predictor configuration can change across bootstrap samples.

Dependent Data: The Moving Block Bootstrap

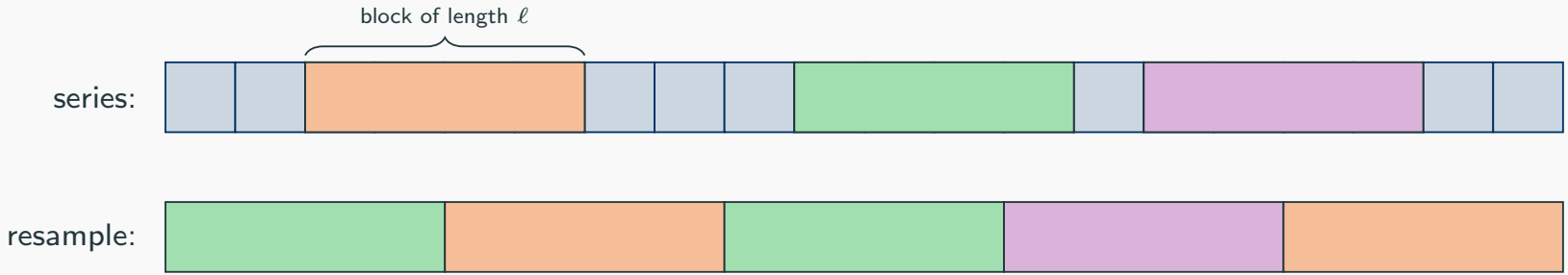
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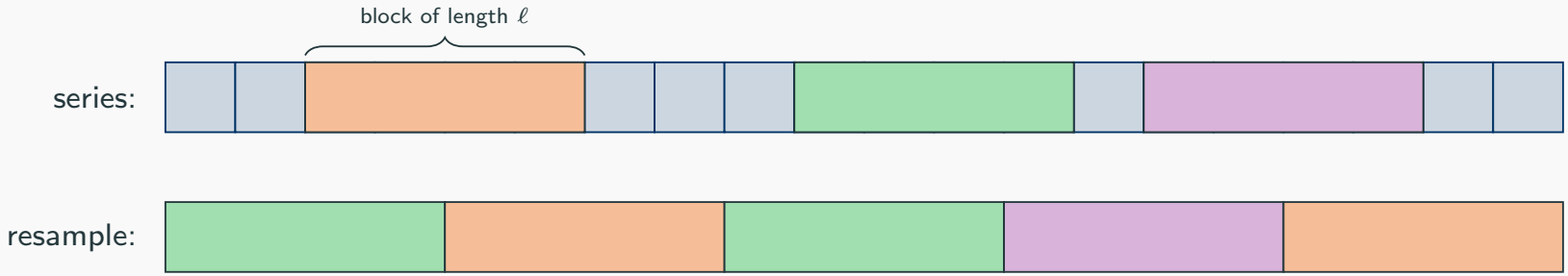
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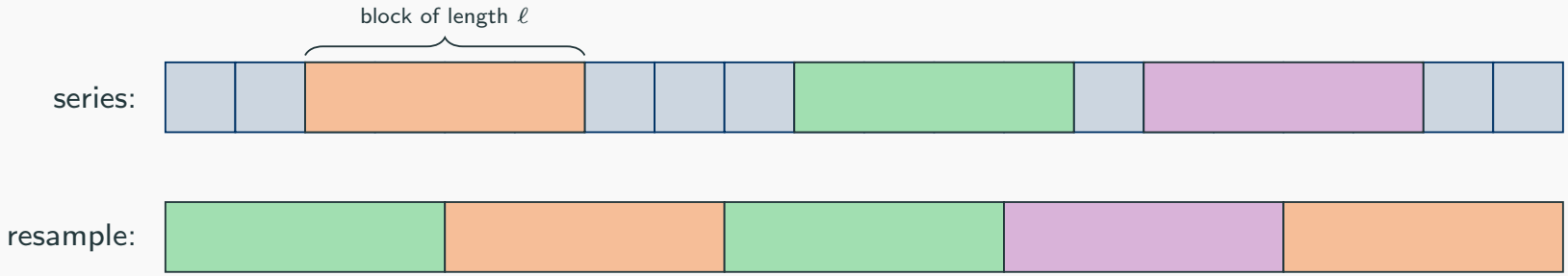
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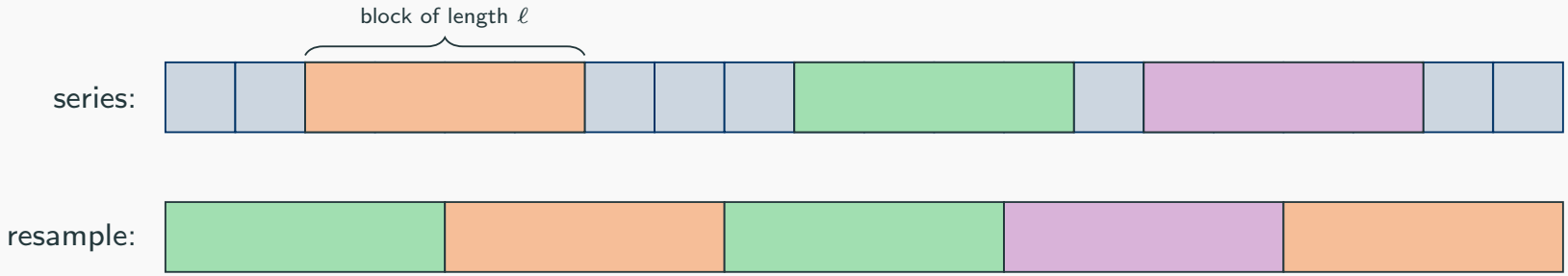
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- **Climate:** trends in monsoon rainfall need block (e.g. by season) resampling – an i.i.d. bootstrap gives overconfident intervals.

Which Bootstrap When?

Data situation	Bootstrap	Resampling unit
i.i.d., no trusted model	Nonparametric	observations x_i
i.i.d., trusted parametric model, small n	Parametric	fresh draws from $F_{\hat{\theta}}$
Regression, random design / heteroscedastic	Pairs	cases $(\boldsymbol{x}_i, y_i)$
Regression, fixed design, i.i.d. errors	Residual	residuals e_i
Time series / spatial data	Block	blocks of neighbours
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The golden rule

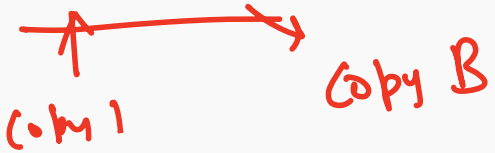
Resample the way the data were generated. The bootstrap replicates whatever randomness you build into the resampling step – and nothing else.

Outline

1. Motivation: Inference When Formulas Run Out
2. The Bootstrap Principle
3. Parametric vs Nonparametric Bootstrap
4. Bootstrap Confidence Intervals
   - Normal and Percentile Intervals
   - The Basic (Pivotal) Interval
   - The Bootstrap-t Interval
   - The BCa Interval
   - Worked Example and Coverage Study
5. The Jackknife (Brief)
6. Cross-Validation
7. Monte Carlo Hypothesis Testing and Permutation Tests
8. Likelihood-Based Inference via Simulation
9. Simulation vs Asymptotic Inference
10. Unit Summary

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- After B resamples we hold a histogram of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$.



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- Write $\hat{\theta}^*_{(\alpha)}$ for its α-quantile (the $(B\alpha)$-th ordered value).

$B = 1000$, $\alpha = 0.05$ $\Rightarrow$ 5th quantile

$\hat{\theta}^*_{(1)} < \hat{\theta}^*_{(2)} < \cdots < \hat{\theta}^*_{(B)}$

$\downarrow$ $B\alpha$ th ordered value.

$1000 \times 0.05 = 50$

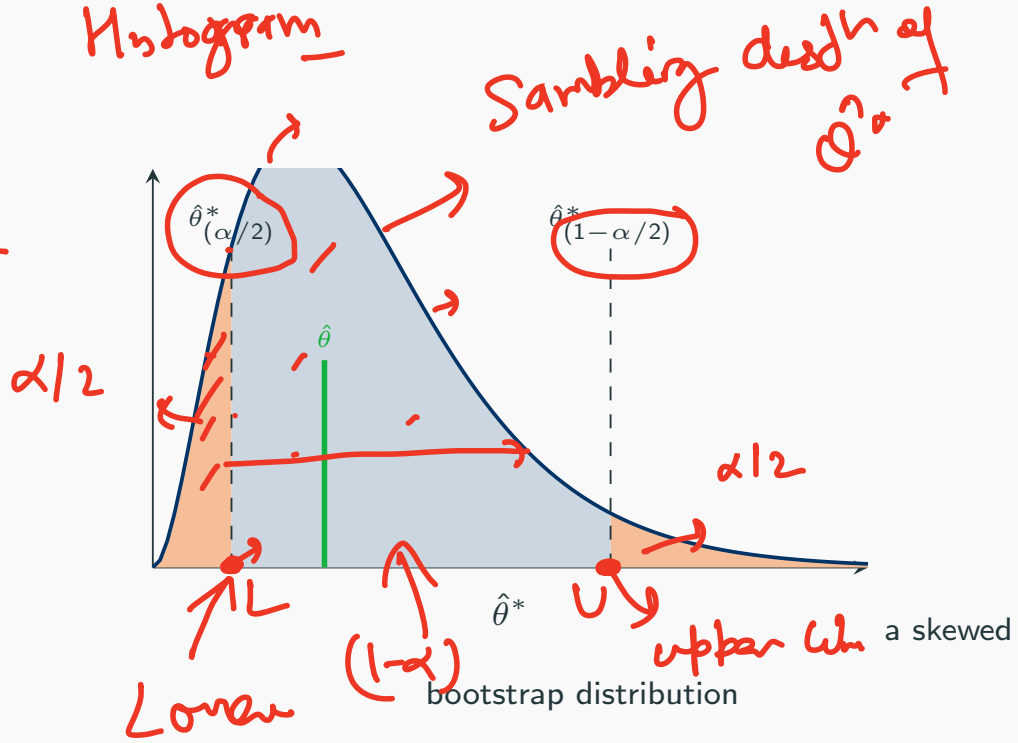
$(1-\alpha)$ % CI(s)

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1. Normal (standard) bootstrap interval

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- It is exactly right if some (unknown) transformation makes $\hat{\theta}$ Normal and unbiased – but it does not correct for bias or skewness on its own.

Normal → Bootstrap

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